

Research Project

Comparison of 2D and 3D Implementation of a Streamlined Body in Eddy-Dominant Flow

Institut für Wasserbau und Technische Hydromechanik, Professur Wasserbau

Presented by:

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ABS 04-001 / May 13, 2024

Motivation

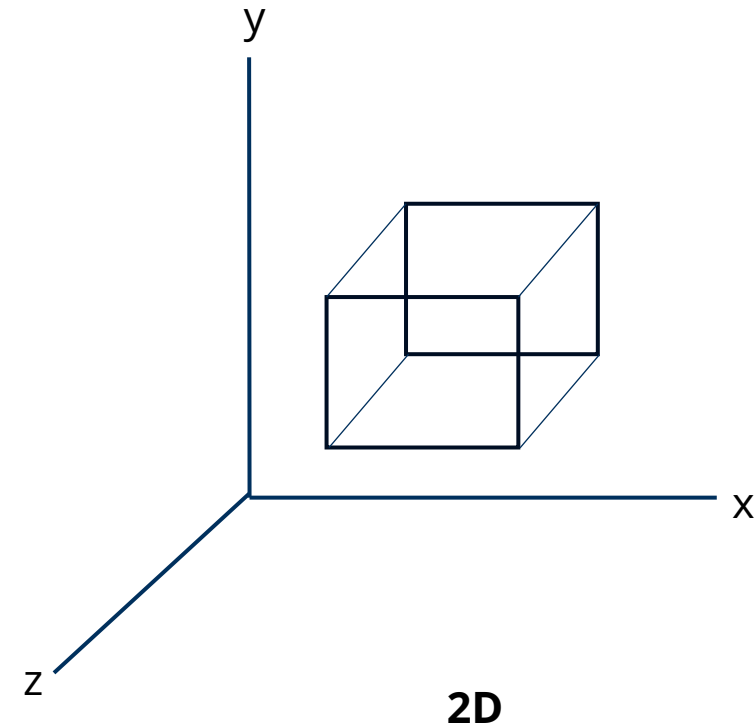
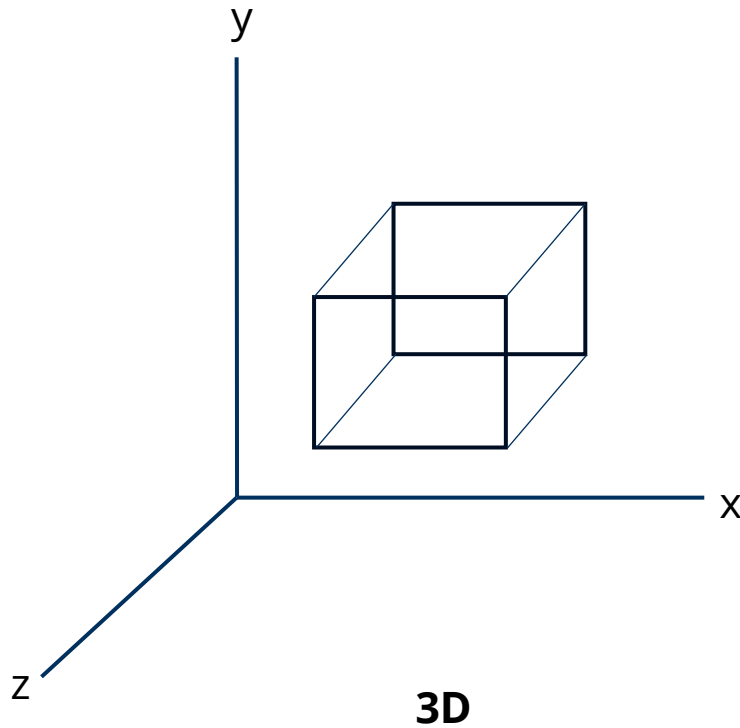
- Aquatic fauna face various disturbances.
- Important to know the flow characteristics and to predict them.
- Project aims to establish a comparison between 2D, and 3D model.
- Numerical Simulation employed for comparison.



[Source](#)

Motivation

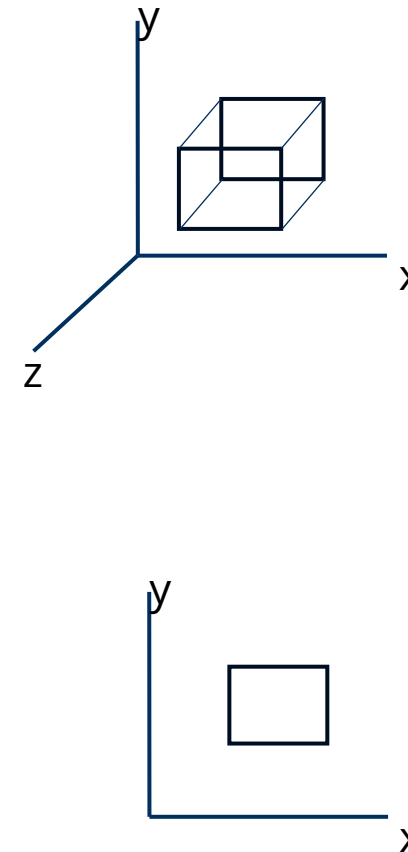
- Model simplification



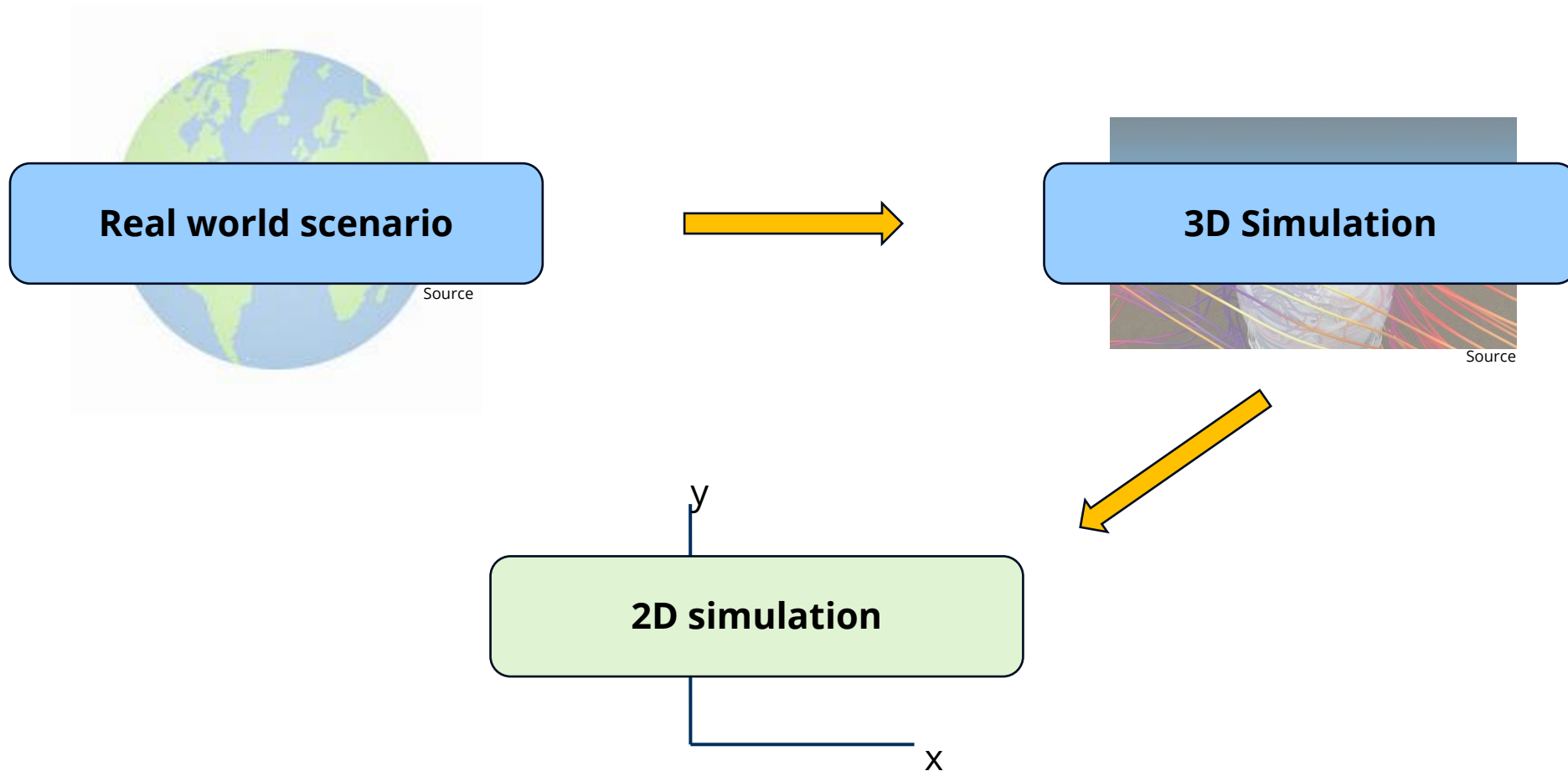
Motivation

- Model simplification
- Dimensionality effects
- Computational cost
 - Lower cells
 - Faster meshing
 - Faster simulation

Disclaimer: A 2D model cannot depict reality !



Motivation



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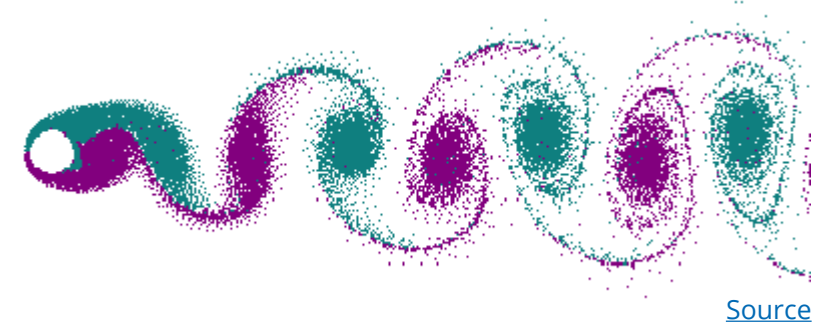
Conclusion

Introduction

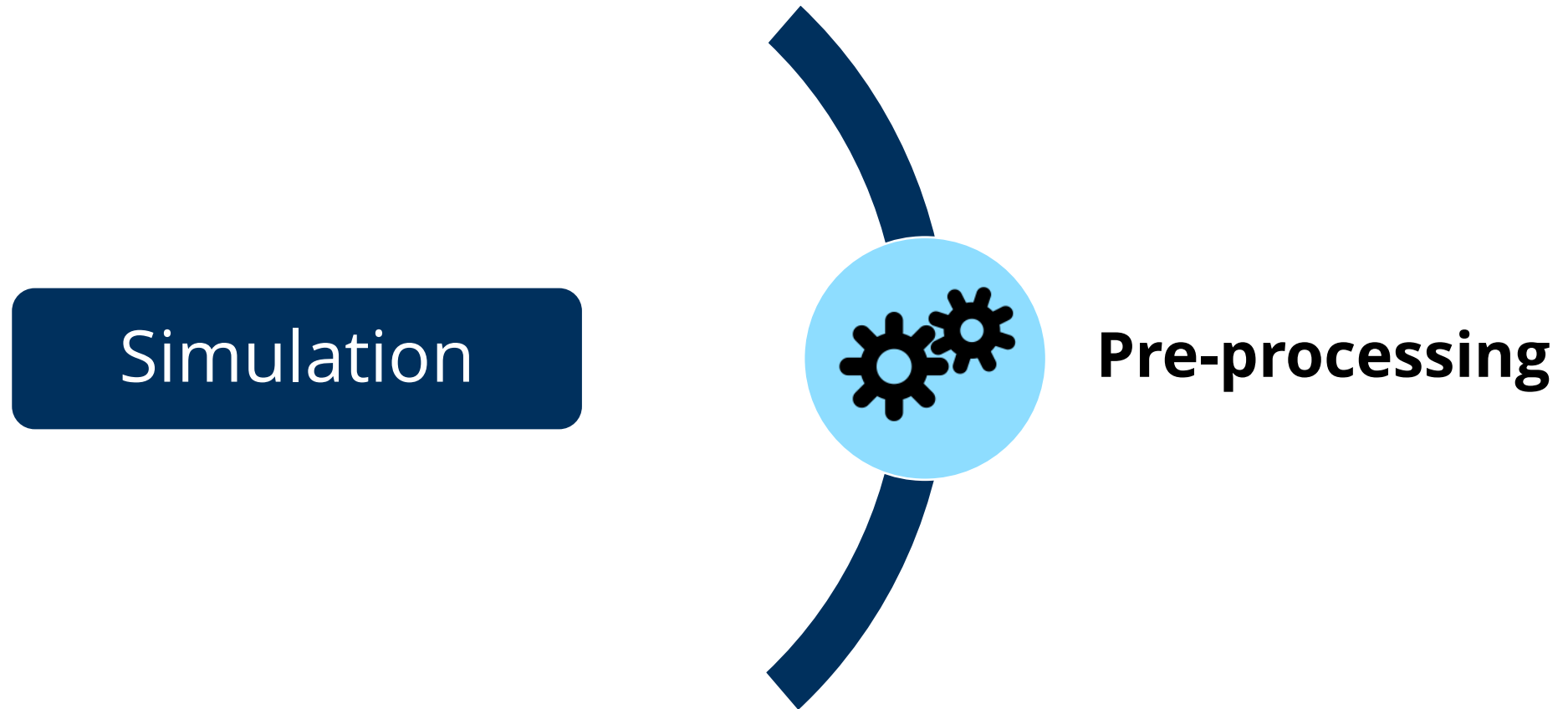
Background

- Kármán vortex street is a repeating pattern of swirling vortices, caused by a process known as vortex shedding.
- Vortex street forms only at a certain range of flow velocities characterised by the Reynolds number.
- Analysed using Strouhal number.

$$St = 0.198(1 - \frac{19.7}{Re})$$



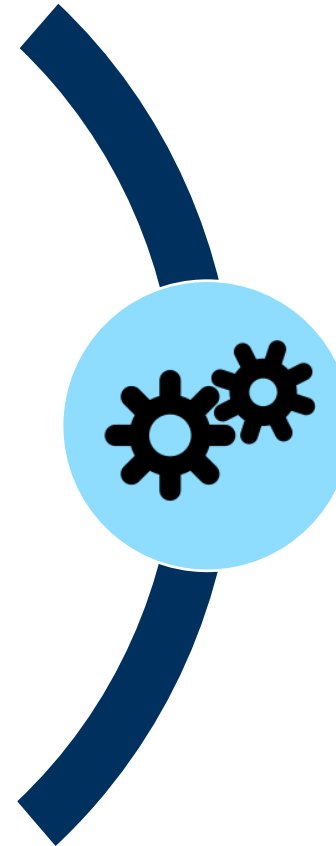
Methodology



Methodology

Simulation

- Geometry Preparation
- Meshing
- Boundary Conditions
- Initial Conditions

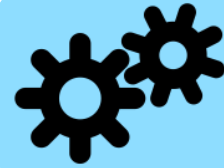


Pre-processing

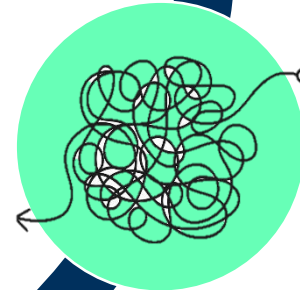
Methodology

Simulation

- Selection of Solver
- Turbulence Model
- Convergence Criteria
- Stopping Criteria



Pre-processing

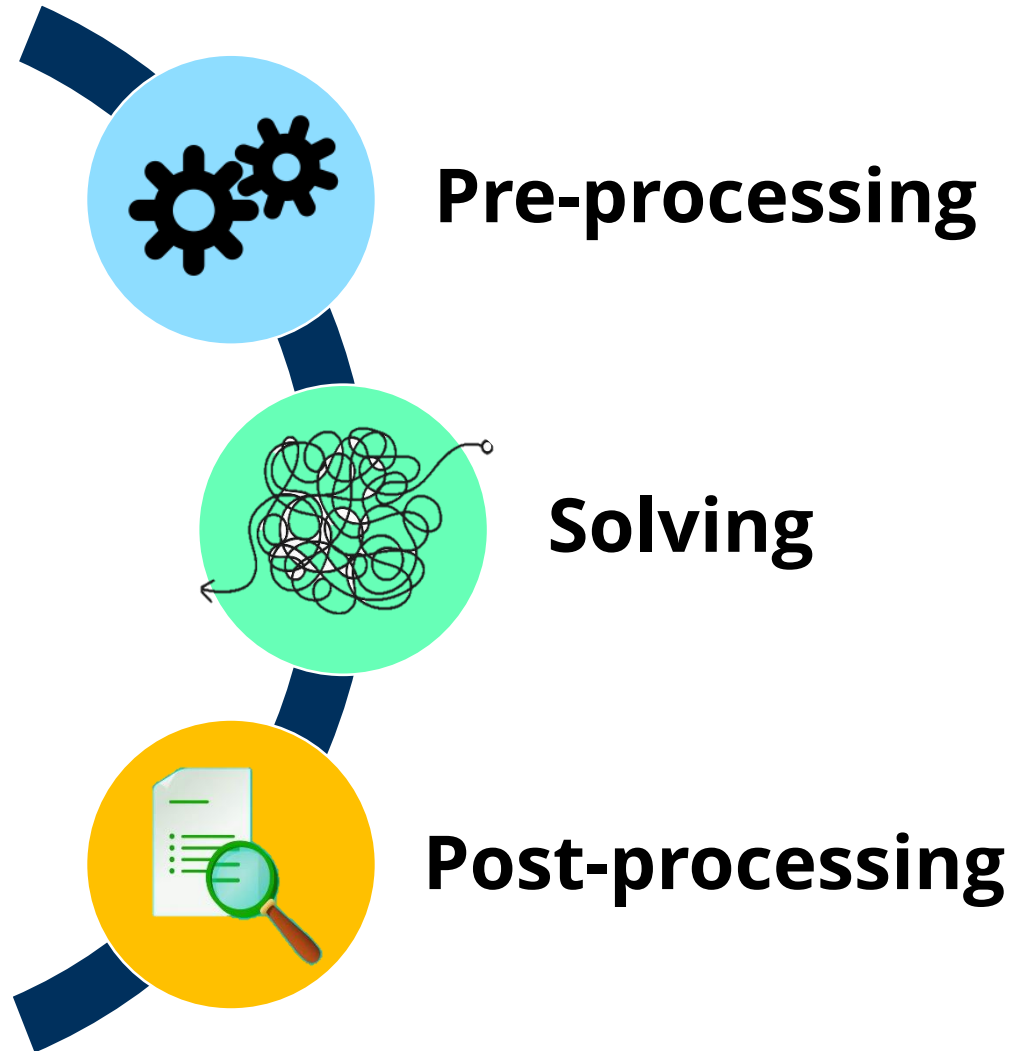


Solving

Methodology

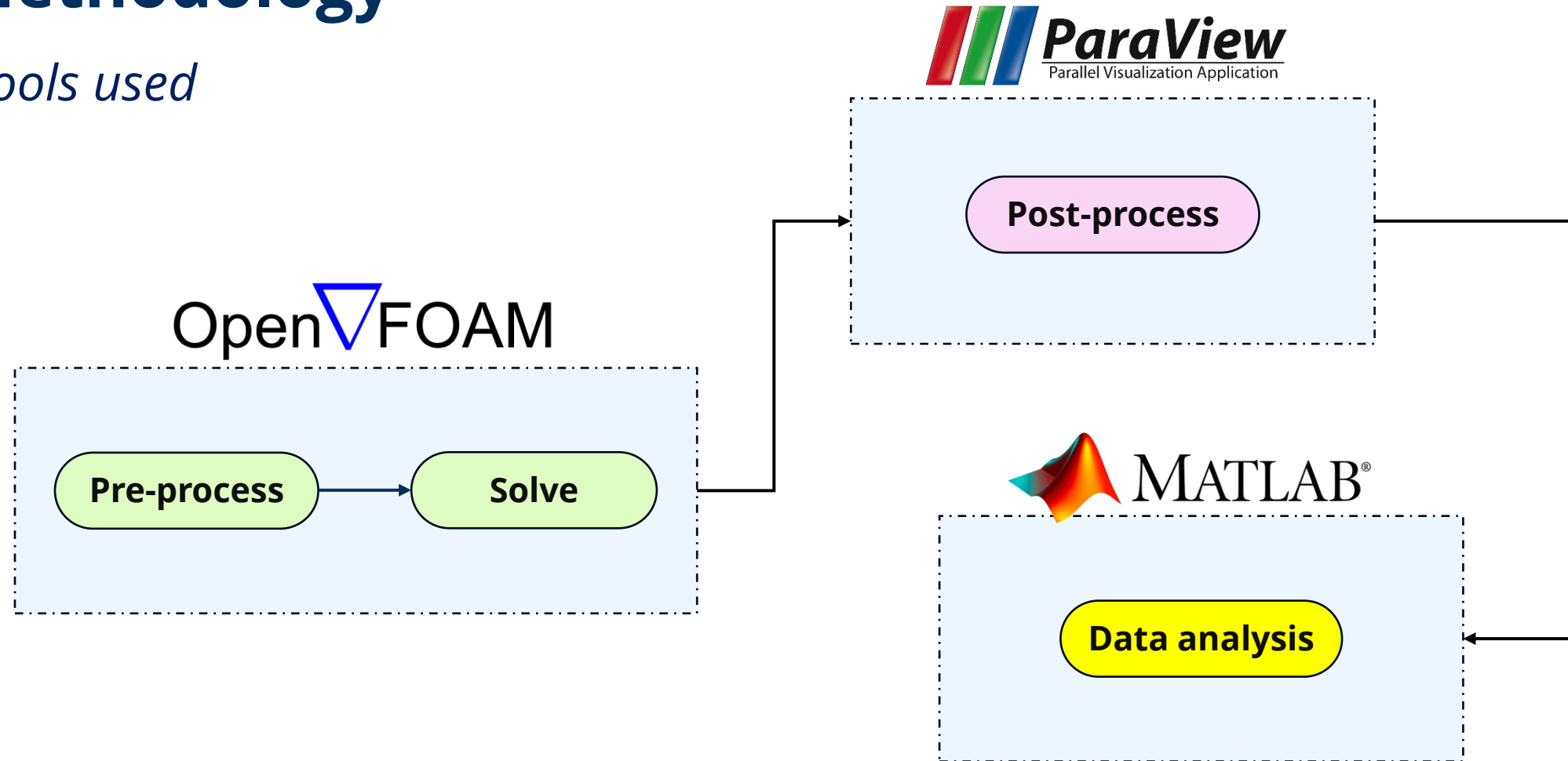
Simulation

- Data Extraction
- Data Visualization
- Quantitative Analysis
- Validation



Methodology

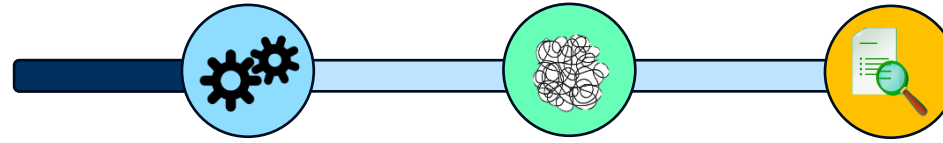
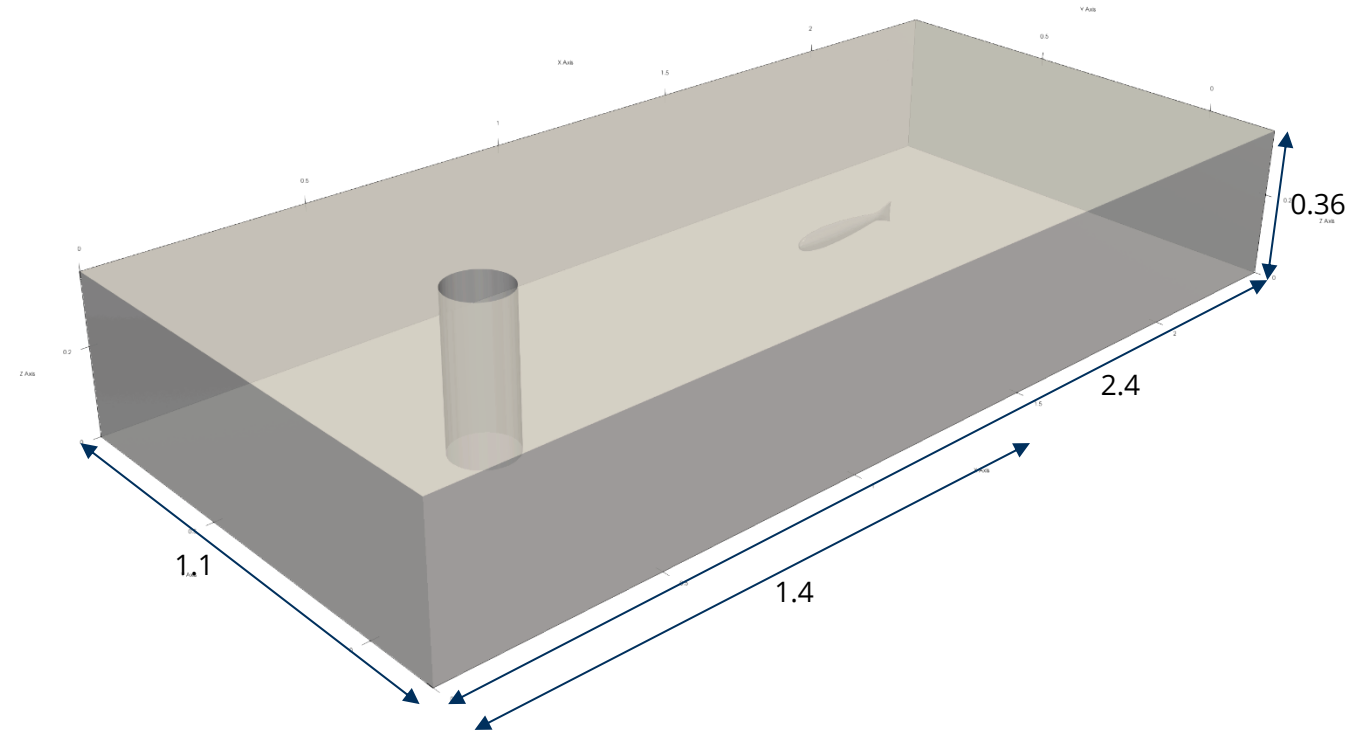
Tools used



Methodology

Pre-process : Geometry

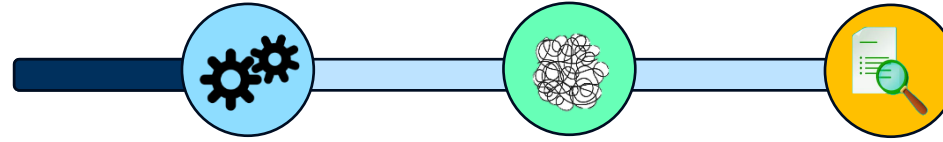
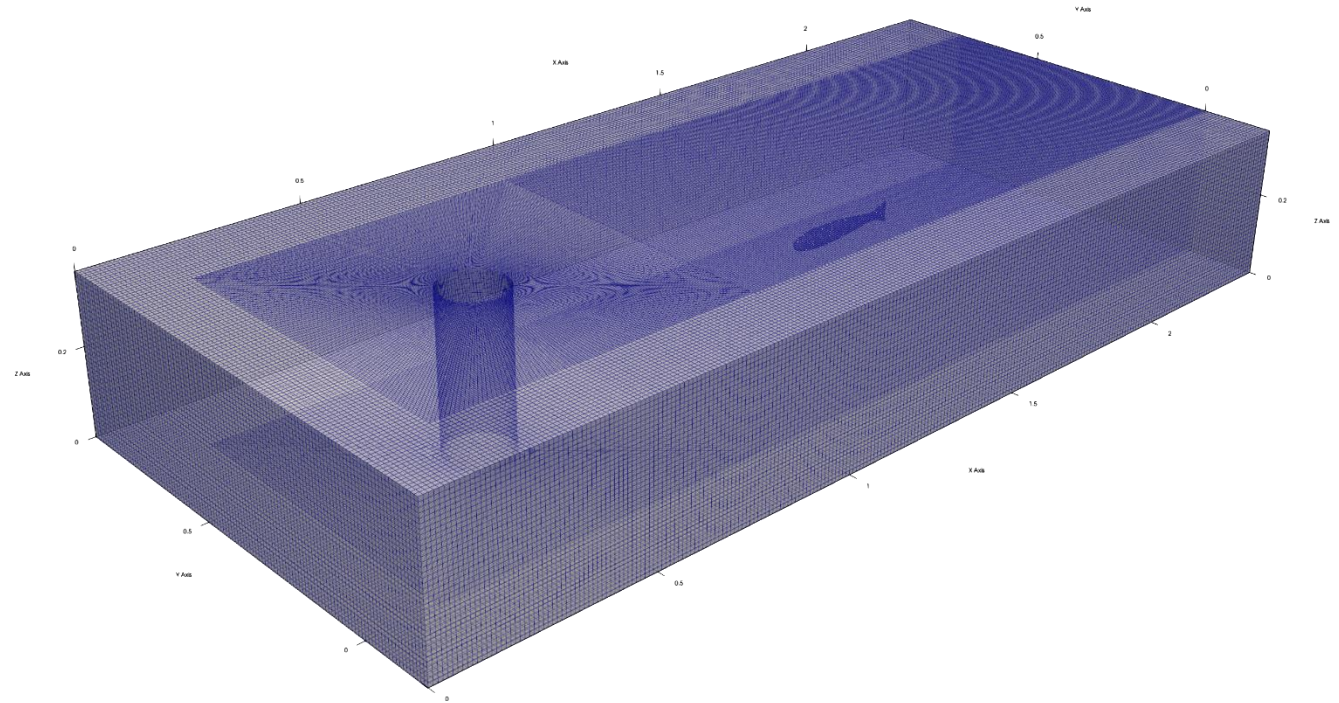
- Domain : 2.4m x 1.1m x 0.36m (l x b x h).
- Cylindrical structure (0.15m).
- Fish structure placed in vortex domain.



Methodology

Pre-process : Meshing

- Meshing using blockMesh and snappyHexMesh.
- blockMesh used to create structured and hexahedral meshes.
- snappyHexMesh used to generate high-quality, complex geometries meshes.



Methodology

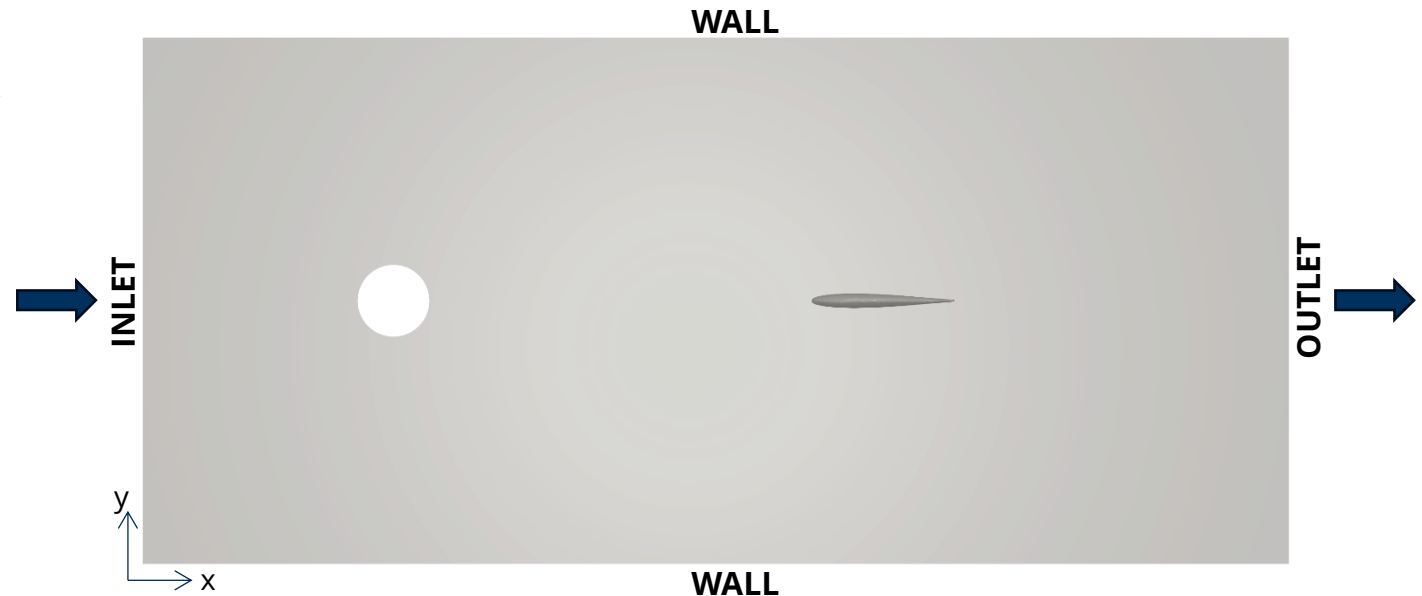
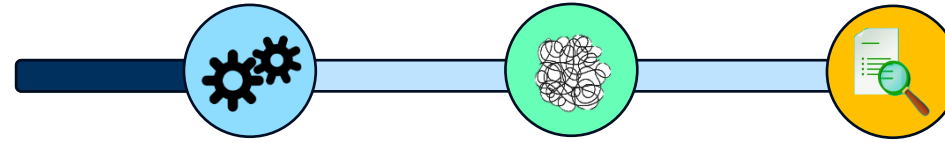
Pre-process : Initialisations

Boundary Conditions:

- Inlet: uniform velocity of 1m/s.
- Outlet: zero gradient pressure.
- Walls: slip
- Cylinder/ Fish: no slip

Governing equation: NS equation.

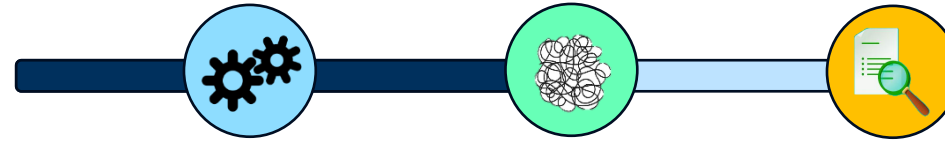
Turbulence model: $k-\omega$ SST.



Methodology

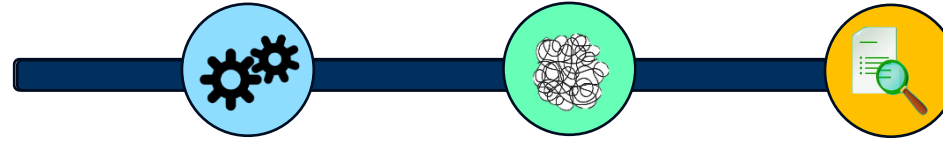
Solve

- Navier-Stokes equation is solved until convergence is achieved.
- Convergence typically influenced by mesh resolution, time step-size and convergence criteria.
- PIMPLE solver: suitable for transient flows.



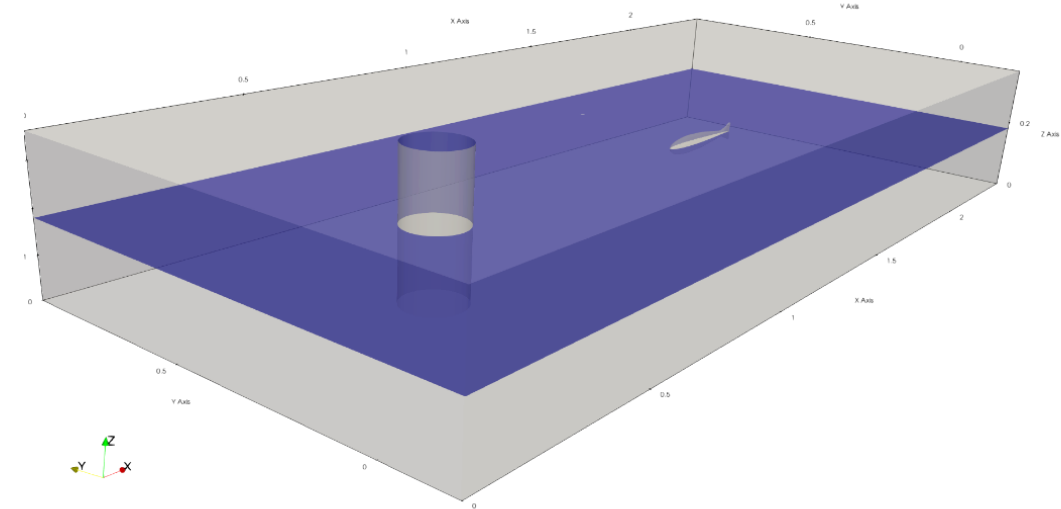
```
martinmatt13@MartinMathew x + v
PIMPLE: iteration 2
smoothSolver: Solving for Ux, Initial residual = 0.00010353131, Final residual = 1.6549797e-07, No Iterations 1
smoothSolver: Solving for Uy, Initial residual = 0.0001649391, Final residual = 2.9688343e-07, No Iterations 1
GAMG: Solving for p, Initial residual = 0.2497767, Final residual = 0.0010410084, No Iterations 2
GAMG: Solving for p, Initial residual = 0.0018790108, Final residual = 7.838129e-06, No Iterations 4
GAMG: Solving for p, Initial residual = 0.000266993, Final residual = 2.1284937e-06, No Iterations 5
time step continuity errors : sum local = 5.4064175e-11, global = 3.7279932e-19, cumulative = 9.9805406e-17
GAMG: Solving for p, Initial residual = 0.13555862, Final residual = 0.0005673176, No Iterations 2
GAMG: Solving for p, Initial residual = 0.0011082381, Final residual = 4.4493125e-06, No Iterations 4
GAMG: Solving for p, Initial residual = 0.00015711659, Final residual = 5.30767281e-07, No Iterations 6
time step continuity errors : sum local = 1.359525e-11, global = 2.0304599e-19, cumulative = 1.0000845e-16
PIMPLE: iteration 3
smoothSolver: Solving for Ux, Initial residual = 8.5102541e-05, Final residual = 1.2858763e-07, No Iterations 1
smoothSolver: Solving for Uy, Initial residual = 0.00013623281, Final residual = 2.2554531e-07, No Iterations 1
GAMG: Solving for p, Initial residual = 0.12320606, Final residual = 0.00054960928, No Iterations 2
GAMG: Solving for p, Initial residual = 0.00099444738, Final residual = 9.0666233e-06, No Iterations 3
GAMG: Solving for p, Initial residual = 0.00014441423, Final residual = 1.252875e-06, No Iterations 5
time step continuity errors : sum local = 3.0893448e-11, global = 3.5630751e-19, cumulative = 1.0036476e-16
GAMG: Solving for p, Initial residual = 0.070222577, Final residual = 0.00031515262, No Iterations 2
GAMG: Solving for p, Initial residual = 0.00058875921, Final residual = 5.4812288e-06, No Iterations 3
GAMG: Solving for p, Initial residual = 8.6240322e-05, Final residual = 7.4995542e-07, No Iterations 5
time step continuity errors : sum local = 1.8553239e-11, global = 3.8721681e-19, cumulative = 1.0075198e-16
PIMPLE: iteration 4
smoothSolver: Solving for Ux, Initial residual = 6.4068471e-05, Final residual = 9.3116651e-08, No Iterations 1
smoothSolver: Solving for Uy, Initial residual = 0.00010343874, Final residual = 1.6380692e-07, No Iterations 1
GAMG: Solving for p, Initial residual = 0.062285752, Final residual = 0.00034305515, No Iterations 2
GAMG: Solving for p, Initial residual = 0.00072278153, Final residual = 4.3608732e-06, No Iterations 4
GAMG: Solving for p, Initial residual = 0.00011917247, Final residual = 9.2906978e-07, No Iterations 5
time step continuity errors : sum local = 2.3456884e-11, global = 1.9479513e-19, cumulative = 1.0094677e-16
GAMG: Solving for p, Initial residual = 0.036375831, Final residual = 0.00020025934, No Iterations 2
GAMG: Solving for p, Initial residual = 0.00042023063, Final residual = 2.4934464e-06, No Iterations 4
GAMG: Solving for p, Initial residual = 6.935981e-05, Final residual = 5.3431796e-07, No Iterations 5
time step continuity errors : sum local = 1.3139959e-11, global = 9.1951241e-20, cumulative = 1.0103872e-16
PIMPLE: iteration 5
smoothSolver: Solving for Ux, Initial residual = 4.5172299e-05, Final residual = 6.389653e-08, No Iterations 1
smoothSolver: Solving for Uy, Initial residual = 7.3265326e-05, Final residual = 1.1267083e-07, No Iterations 1
GAMG: Solving for p, Initial residual = 0.038051346, Final residual = 0.00022903324, No Iterations 2
GAMG: Solving for p, Initial residual = 0.0005837318, Final residual = 4.5632077e-06, No Iterations 4
GAMG: Solving for p, Initial residual = 9.9020658e-05, Final residual = 6.9848412e-07, No Iterations 5
time step continuity errors : sum local = 1.7413617e-11, global = 5.0192663e-19, cumulative = 1.0154065e-16
GAMG: Solving for p, Initial residual = 0.022277772, Final residual = 0.00013410374, No Iterations 2
```


Results

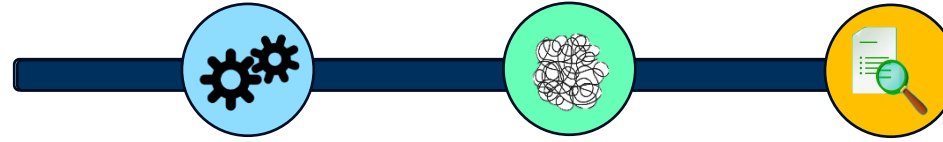


Post-process

- Category: 1. qualitative / visualization based
2. quantitative/ profile, trend based
- Post-process is the basis for comparison.
- Different post-process techniques employed at discrete times for quantitative analysis.



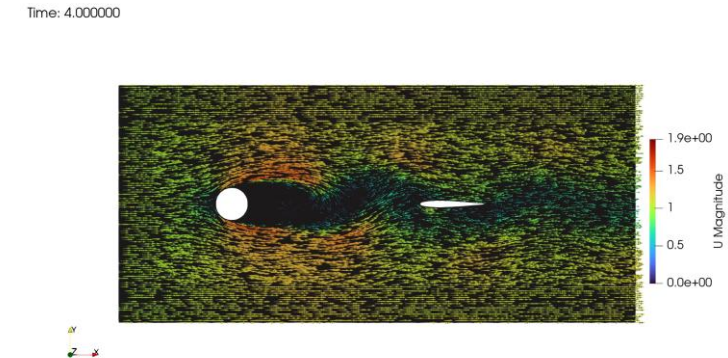
Results



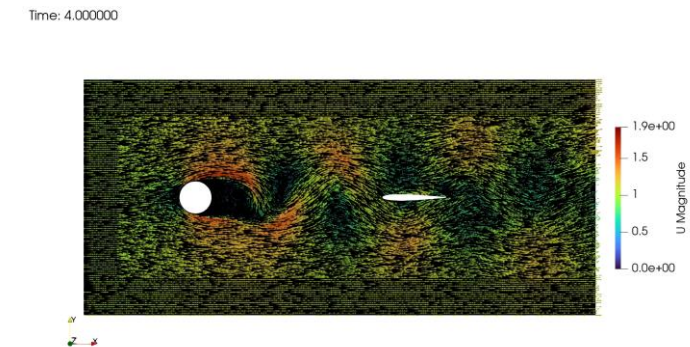
Post-process : qualitative / visualization

- Visual Inspection of the system/ domain – Glyph.
- Domain is suitable for avoiding far-field effects.
- Far-field effects is the behaviour of fluid flow close to the boundaries.

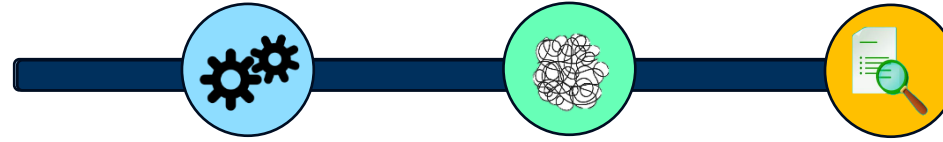
2D



3D



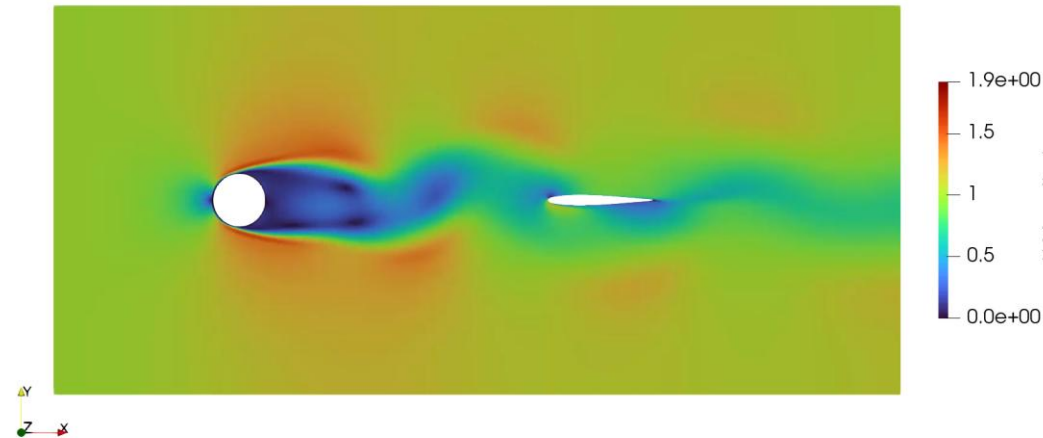
Results



Post-process : qualitative / visualization

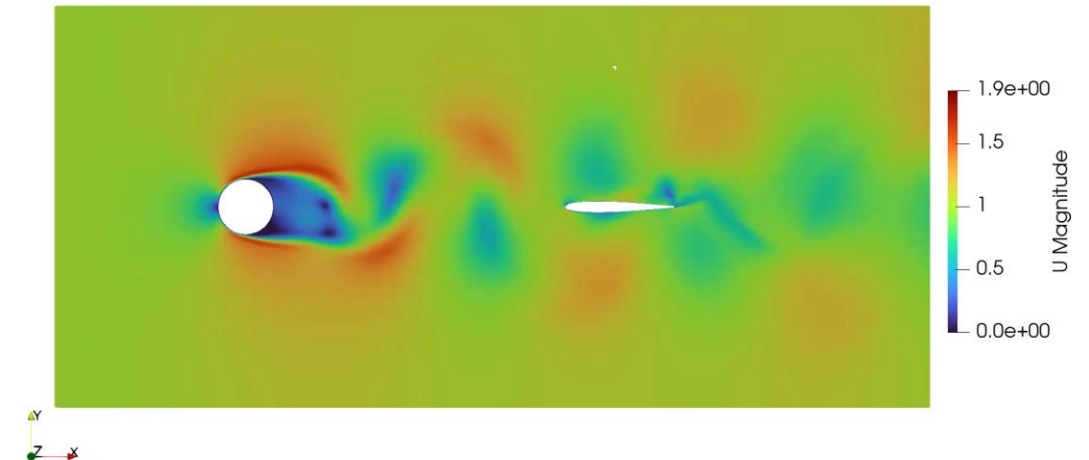
- Visual Inspection of the system/ domain – Velocity pattern.

Time: 4.000000



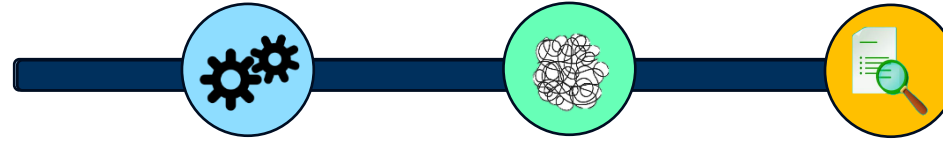
2D

Time: 4.000000



3D

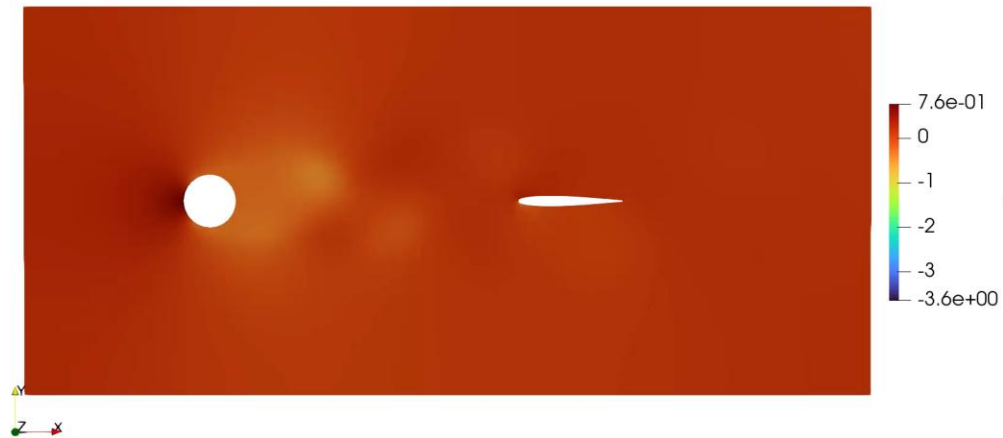
Results



Post-process : qualitative / visualization

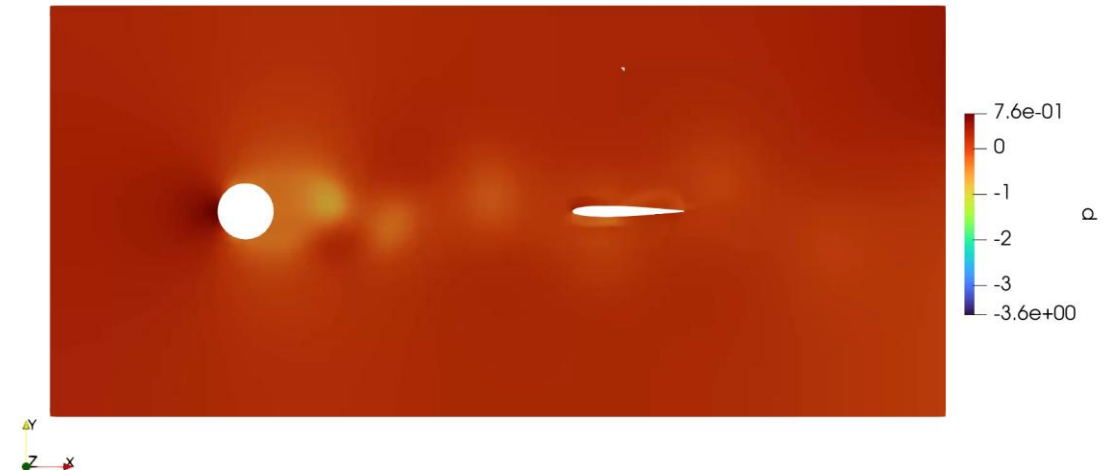
- Visual Inspection of the system/ domain – Pressure pattern.

Time: 4.000000



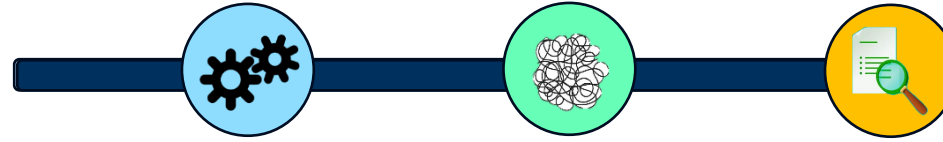
2D

Time: 4.000000



3D

Results



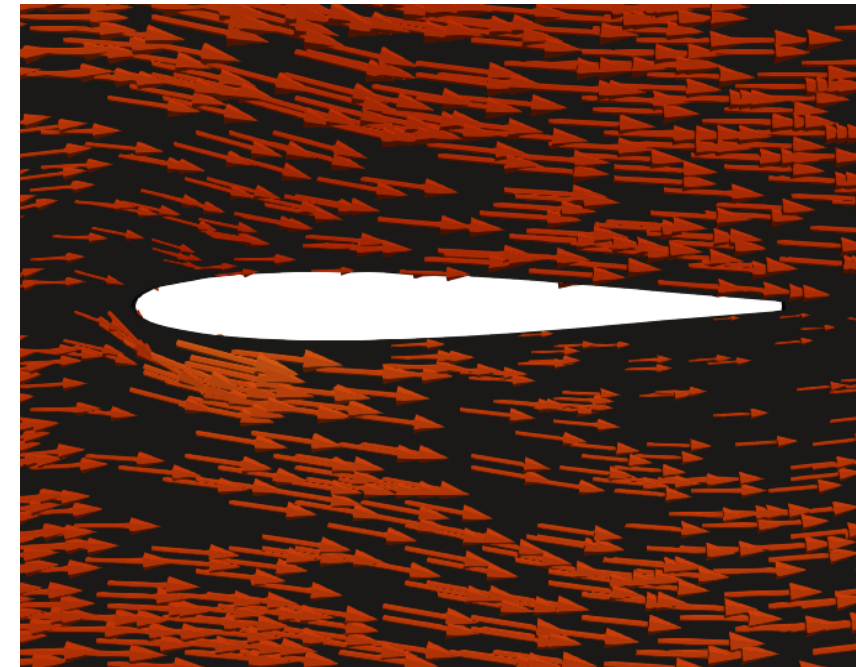
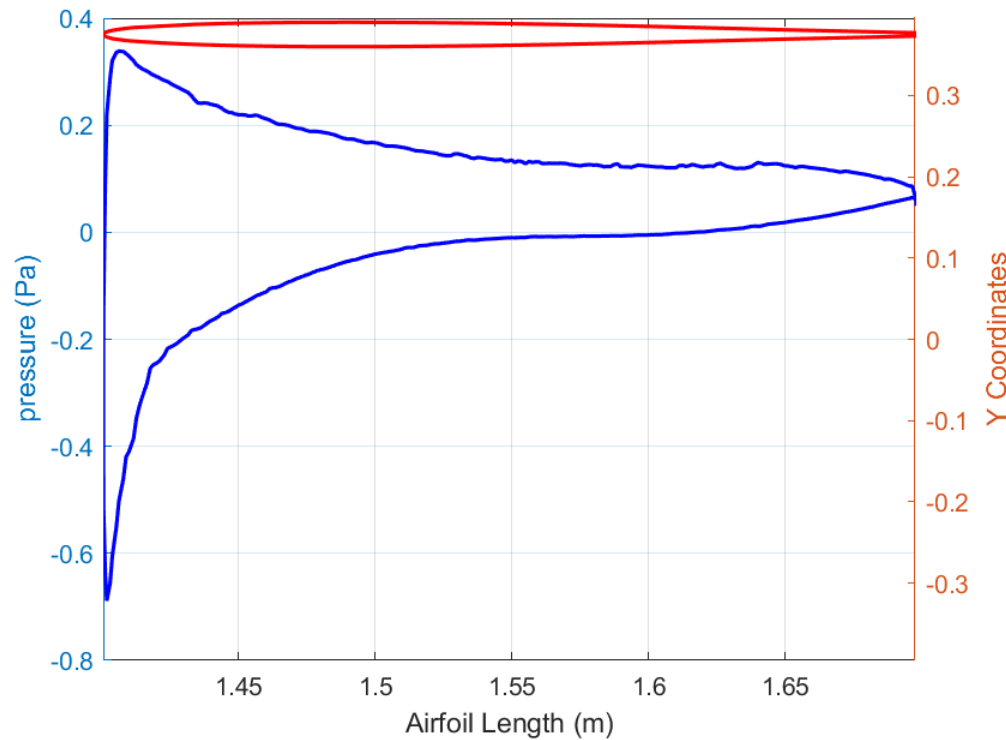
Post-process : quantitative analysis

- Analysis of numerical data generated from the simulation.
- Plot over line functionality to quantify domain.
- Pressure distribution over the fish structure.
- Drag coefficient over the fish structure.

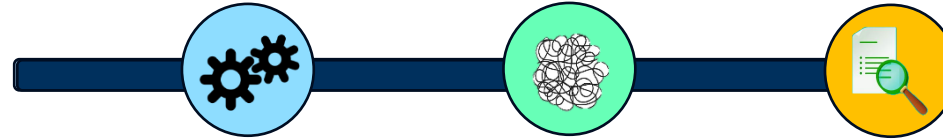
Results

Post-process : quantitative analysis

- Pressure variation at 4 second for 2D.

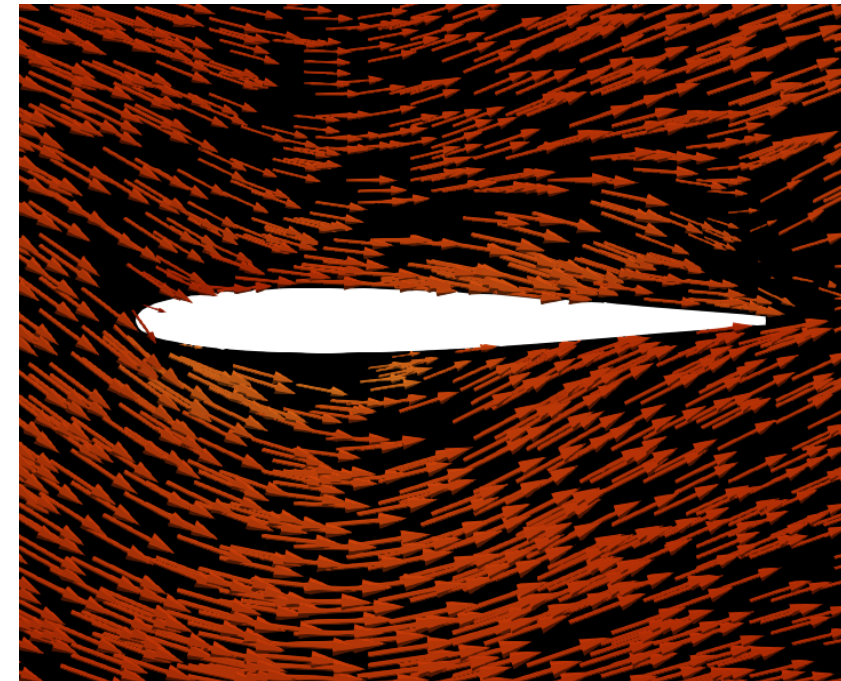
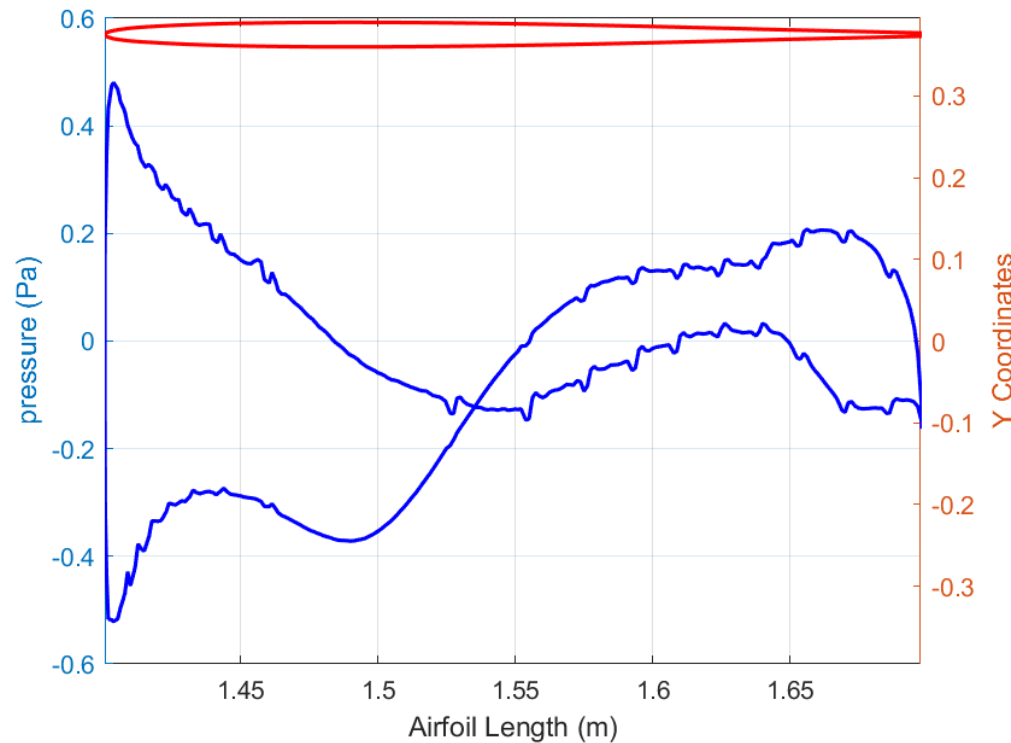


Results



Post-process : quantitative analysis

- Pressure variation at 4 second for 3D.



Conclusion

Inference

- While 2D model offers computational efficiency, 3D model provide a more realistic representation of complex flow phenomena.
- 2D model is suitable for qualitative analysis.
- 2D model may be implemented for dimension independent systems.

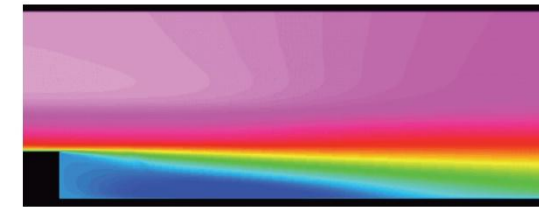
$$St = 0.198(1 - \frac{19.7}{Re})$$

- 3D model is suitable for quantitative analysis (pressure/ drag calculations).

Conclusion

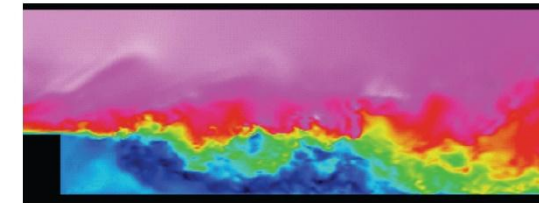
Outlook

- Investigating the effects of different turbulence models (LES).
- Implementation in different numerical solver.
- Studying the impact of varying geometric parameters.
- Experimental validation.



RANS

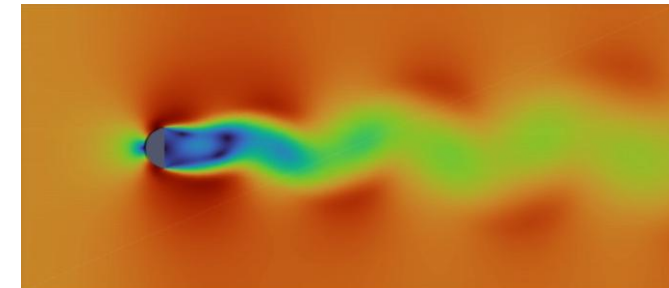
Source: Rémy Fransen, 3rd INCA colloquium, ONERA, Toulouse (2011)



LES

Source: Rémy Fransen, 3rd INCA colloquium, ONERA, Toulouse (2011)

[Source](#)



Acknowledgment



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Thank you